Bounds on Round Skipping Attacks on the SPN Hash functions

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Abstract

Round-skipping attacks reduce the cost of the Constrained-Input Constrained-Output (CICO) problem on SPN permutations by algebraically bypassing a prefix of full rounds. Recent work showed that Poseidon2 with its non-MDS linear layer admits skipping up to $\approx t/4$ rounds, leaving open the question of how many rounds can be skipped when the linear layer is MDS.

We introduce *skipping trails*—truncated differential trails whose arity-2 structure forces a ratio constraint to propagate through the S-box layer—and prove that an m-round skipping trail for a linear layer of branch number b requires $b \leq 2t/m + 2$. For MDS matrices ($b = t + 1$, $t \geq 3$), this immediately gives $m \leq 2t/(t-1) < 3$, ruling out skipping three or more full rounds. We show the bound is tight by exhibiting an explicit $m = 2$ skipping trail of width $k = 1$ for the circulant MDS matrix $\mathrm{circ}(2, 1, 1)$ over $\mathbb{F}_{11}$ and KoalaBear ($p = 2^{31} - 2^{24} + 1$): for any round constants, a center satisfying the trail exists and can be found by solving a single cubic equation.

1 Introduction

Zero-knowledge proof systems—SNARKs, STARKs, and Bulletproofs—have become core infrastructure for privacy-preserving computation. Their inner loops are dominated by hash computations that must be expressed natively over prime fields with as few multiplications as possible. The Poseidon family of hash functions [1, 2], built on the Substitution-Permutation Network (SPN) design, is the dominant answer: deployed in Filecoin, Mina, Plonky3, and many other production systems, it achieves the best known trade-off between algebraic complexity and security.

An SPN permutation operates on a state $W \in \mathbb{F}_p^t$ by alternating a non-linear S-box layer $\mathcal{S}$ (applying the power map $x \mapsto x^d$ to each element) with a linear diffusion layer $\mathcal{L}$ (multiplication by a $t \times t$ matrix M). Security is measured primarily through the *Constrained-Input Constrained-Output* (CICO) problem: given partial constraints on the input and output of the permutation, find a satisfying assignment. CICO hardness underlies collision resistance and preimage resistance, and concretizes the security of the Poseidon Initiative bounty challenges [5].

Round-skipping attacks reduce the CICO problem by algebraically bypassing a prefix of rounds. Merz and Rodríguez García [3] recently showed that the structured linear layers of Poseidon2(b)—optimized for hardware efficiency but not MDS—admit skipping attacks that bypass up to $\approx t/4$ full rounds by spending roughly four degrees of freedom per skipped round. This raises a fundamental open question: *how many rounds can be skipped when the linear layer is an MDS matrix?* Poseidon1 uses MDS matrices (via the Cauchy or circulant construction, achieving branch number $b = t + 1$), so this question directly bounds the attack surface of the Poseidon Initiative bounty instances.

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Our Contributions

- **Skipping trail framework (Section 3).** We introduce *skipping trails*: truncated differential trails satisfying an algebraic compatibility condition that forces the ratio of active state element pairs to propagate linearly through the S-box layer. A valid m-round skipping trail certifies that a structured family of inputs maps through m rounds to a predictable output set, enabling the adversary to reduce a CICO instance on the full permutation to one on the remaining $R_F - m$ full rounds and R_P partial rounds.

- **Tight bounds on round skipping (Section 4).** We prove that an m-round skipping trail can exist only if the branch number b of $\mathcal{L}$ satisfies

$$b \leq \frac{2t}{m} + 2.$$

  For MDS matrices ($b = t+1$, $t \geq 3$), this gives $m \leq 2t/(t-1) < 3$, ruling out skipping $m \geq 3$ rounds. The bound is tight: $m = 2$ is achieved (Section 5).

- **Concrete trails (Section 5).** We exhibit an explicit tight $m = 2$ skipping trail of width $k = 1$ for the circulant MDS matrix $M = \mathrm{circ}(2, 1, 1)$ over $\mathbb{F}_{11}$ (and over KoalaBear), showing that the bound $m \leq 2$ of Proposition 2 is achieved for $t = 3$, $d = 3$. The trail is based on an eigenvector of M and its existence is proved for arbitrary round constants.

- **Complexity analysis and implementation (Section ??).** We quantify the CICO cost reduction as a function of the trail width and R_P. We provide an implementation that finds skipping trails for concrete Poseidon1-KoalaBear matrices and demonstrates the round-skipping attack on the bounty instances.

Scope. We analyze skipping within the full-round layer only ($m \leq R_F$). Extending skipping into the partial-round layer is qualitatively different and is left for future work.

Organization. Section 2 covers SPN notation, Poseidon1, MDS matrices, and the CICO problem. Section 3 defines truncated differential trails and skipping trails. Section 4 proves the main bounds. Section 5 presents concrete $m = 2$ skipping trails for the circulant MDS matrix $\mathrm{circ}(2, 1, 1)$ over $\mathbb{F}_{11}$ and KoalaBear. Section **??** describes the implementation and complexity analysis.

1.1 Related Work

Poseidon hash functions. Poseidon1 [1] (Grassi, Khovratovich, Rechberger, Roy, Schofnegger, 2021) is an SPN hash function over $\mathbb{F}_p$ with R_F *full rounds* (all t S-boxes active) flanking R_P *partial rounds* (one S-box active), and an MDS linear layer with branch number $b = t + 1$. Poseidon2 [2] replaces the MDS matrix with a structured matrix achieving better circuit efficiency; this matrix is not MDS in general, and its reduced diffusion is the structural weakness exploited in [3].

Round-skipping attacks. The idea of bypassing SPN rounds algebraically has roots in the *rebound attack* of Mendel, Rechberger, Schläffer, and Thomsen [4] on AES-based hash functions. In the rebound attack, the adversary satisfies differential constraints through the middle rounds by working inward from both sides simultaneously; the adversary controls the degrees of freedom through the linear (MDS) layer and the non-linear S-box. Merz and Rodríguez García [3] adapted this idea to Poseidon2 by identifying truncated differentials with a ratio-preserving property through

the power-map S-box. Their attack skips up to $\approx t/4$ rounds and yields the first algebraic collision attack that outperforms the corresponding preimage attack on Poseidon2. They explicitly leave open the question of how far round skipping can go for MDS-based SPNs; our paper answers this question.

Algebraic attacks. Algebraic attacks on Poseidon-type primitives include Gröbner-basis methods, linearization, and interpolation. These complement round-skipping attacks: our results show that, for MDS matrices, at most 3 full rounds can be skipped (with $m \geq 4$ being provably impossible), bounding the scope of the round-reduction phase in any hybrid attack.

MDS matrices and branch numbers. The branch number $b = \min_{x \neq 0}(\mathrm{wt}(x) + \mathrm{wt}(Mx))$ quantifies the diffusion of a linear layer, where wt denotes Hamming weight. MDS matrices achieve $b = t + 1$, the maximum by the Singleton bound. The Cauchy construction $M[i][j] = (x_i - y_j)^{-1}$ yields an MDS matrix over any prime field with $p > 2t$; circulant MDS matrices, used in Plonky3 and the Poseidon Initiative bounty instances, achieve the same branch number with a more regular structure. The distinction between the two constructions becomes relevant in Section 5.

2 Preliminaries and Notation

2.1 SPN Permutations

Let $\mathbb{F} = \mathbb{F}_p$ be a prime field and $t \geq 2$ an integer. An *SPN permutation* $\mathcal{P}$ operates on the state $W \in \mathbb{F}^t$ as a composition of R rounds. Each round i applies three operations in sequence:

1. $\mathcal{L}$ — a linear transformation that multiplies the state by a fixed matrix $M \in \mathbb{F}^{t \times t}$;
2. $\mathcal{S}$ — a non-linear substitution layer that applies the power map $x \mapsto x^d$ coordinate-wise;
3. $\mathcal{C}_i$ — a constant-addition layer that adds a round-specific vector of constants to the state.

We write the i-th round as

$$W_i^{(A)} \xrightarrow{\mathcal{L}} W_i^{(B)} \xrightarrow{\mathcal{S}} W_i^{(C)} \xrightarrow{\mathcal{C}_i} W_{i+1}^{(A)}.$$

Here $W_i^{(A)}$ is the state at the beginning of round i (after constants of round $i-1$), $W_i^{(B)}$ is the state after the linear layer, and $W_i^{(C)}$ is the state after the S-box layer.

2.2 The CICO Problem

The primary security notion for SPN-based hash functions is the *Constrained-Input Constrained-Output* (CICO) problem.

Definition 1 (CICO)**.** *Given an SPN permutation $\mathcal{P} : \mathbb{F}^t \to \mathbb{F}^t$ and affine constraints*

$$\mathcal{C}_{\text{in}} \subseteq \mathbb{F}^t, \quad \mathcal{C}_{\text{out}} \subseteq \mathbb{F}^t,$$

find $W \in \mathcal{C}_{\text{in}}$ such that $\mathcal{P}(W) \in \mathcal{C}_{\text{out}}$, or determine that no such W exists.

The CICO problem underlies collision and preimage resistance of Poseidon-based hash functions. Concretely, fixing the capacity elements of the input and output yields a collision; fixing the output fully yields a preimage. The Poseidon Initiative bounty challenges [5] are cast as CICO instances.

2.3 Poseidon1

Poseidon1 [1] is an SPN permutation over $\mathbb{F}_p^t$ with a *mixed-round* structure: it consists of R_F *full rounds*, in which the S-box layer is applied to all t state elements, flanking R_P *partial rounds*, in which only one designated element is cubed. Concretely, the round schedule is $R_F/2$ full rounds, followed by R_P partial rounds, followed by $R_F/2$ full rounds; the total number of rounds is $R = R_F + R_P$.

The canonical bounty instance over the KoalaBear field uses the parameters

$$p = 2^{31} - 2^{24} + 1, \quad d = 3, \quad t = 16, \quad R_F = 8, \quad R_P = 20.$$

Scope. This paper analyzes round skipping within the full-round layers only ($m \leq R_F$). Extending skipping into the partial-round layer is qualitatively different and is left for future work.

2.4 MDS Matrices and Branch Number

Definition 2 (Branch number)**.** *The* branch number *of a linear map $\mathcal{L}$ with matrix M is*

$$b(M) = \min_{x \neq 0} \big(\mathrm{wt}(x) + \mathrm{wt}(Mx)\big),$$

where $\mathrm{wt}(v)$ *denotes the Hamming weight (number of non-zero coordinates) of $v \in \mathbb{F}^t$.*

Definition 3 (MDS matrix)**.** *A matrix $M \in \mathbb{F}^{t \times t}$ is* MDS *(Maximum Distance Separable) if $b(M) = t + 1$, the maximum allowed by the Singleton bound.*

An equivalent characterization: M is MDS if and only if every square submatrix of M is non-singular.

Two MDS constructions appear in deployed instances of Poseidon1.

- **Cauchy MDS.** For distinct elements $x_1, \ldots, x_t, y_1, \ldots, y_t \in \mathbb{F}_p$ with $x_i \neq y_j$ for all i, j, the matrix $M[i][j] = (x_i - y_j)^{-1}$ is MDS over any prime field $p > 2t$.
- **Circulant MDS.** A circulant matrix M defined by its first row $(c_0, c_1, \ldots, c_{t-1})$ with $M[i][j] = c_{(j-i) \bmod t}$ is MDS for specific choices of $(c_0, \ldots, c_{t-1})$ depending on p and t. The Poseidon Initiative bounty instances and Plonky3 use circulant MDS matrices.

Both constructions achieve $b = t + 1$; the distinction becomes relevant in Section 5.

3 Truncated Differentials and Skipping Trails

A round-skipping attack exploits the observation that a structured family of inputs —an affine coset—can be propagated through a sequence of SPN rounds while remaining an affine coset at the output. The S-box layer is normally the obstacle: it maps an affine family to a non-affine image. Our skipping trail framework identifies the precise algebraic condition on the differential matrices and the center of the coset that forces the S-box to collapse the coset back to a new affine family. We build up the formalism in three layers: propagation through the linear layer (truncated differential), propagation through the S-box (S-compatibility), and the full multi-round structure (skipping trail).

Definition 4. *Let $1 \le k \le t$. A set $\mathcal{W} \subseteq \mathbb{F}^t$* satisfies the matrix differential $\mathbf{A}$*, centered at $W_0 \in \mathbb{F}^t$, if $\mathbf{A} \in \mathbb{F}^{t\times k}$ and*

$$\mathcal{W} = \{\, W_0 + \mathbf{A} \cdot \mathbf{X} : \mathbf{X} \in \mathbb{F}^k \,\}.$$

We call $\mathcal{W}$ a differential coset *of $\mathbf{A}$; the integer k is the* width *of the differential and* $\dim(\mathcal{W}) = \mathrm{rank}(\mathbf{A})$.

Remark 1. *The matrix $\mathbf{A} \in \mathbb{F}^{t\times k}$ plays two roles: its* row support *(the set of indices j with a non-zero j-th row) determines which state coordinates are free, while the rank of $\mathbf{A}$ determines the dimension of the affine subspace. Taking $k = t$ and $\mathbf{A}$ full-rank recovers the whole state space; taking $k = 1$ gives an affine line.*

Definition 5. *A* truncated differential *of width k is a pair $(\mathbf{A}, \mathbf{B})$ with $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{t\times k}$ for some $1 \le k \le t$ such that*

$$\mathbf{A} \xrightarrow{\mathcal{L}} \mathbf{B}, \quad \textit{i.e.}\ \mathrm{Im}(M\mathbf{A}) = \mathrm{Im}(\mathbf{B}),$$

where M is the matrix of $\mathcal{L}$. The indices j with a non-zero j-th row of $\mathbf{B}$ are called the active positions.

Remark 2. *The condition $\mathrm{Im}(M\mathbf{A}) = \mathrm{Im}(\mathbf{B})$ means that both $M\mathbf{A}$ and $\mathbf{B}$ span the same $\mathrm{rank}(\mathbf{A})$-dimensional coordinate subspace of $\mathbb{F}^t$. For an MDS matrix M, the active positions of $\mathbf{A}$ (non-zero rows) and of $\mathbf{B}$ must have the same count, both equal to $\mathrm{rank}(\mathbf{A})$. When $\mathbf{B}$ is chosen to be diagonal this recovers the classical formulation; the general $\mathbb{F}^{t\times k}$ form allows a basis for the output subspace that need not be aligned with coordinate axes.*

Definition 6. *A matrix $\mathbf{B} \in \mathbb{F}^{t\times k}$ is $\mathcal{S}$-*compatible *with a matrix $\mathbf{A}' \in \mathbb{F}^{t\times k'}$ if there exist differential cosets $\mathcal{W}^{(B)}$ of $\mathbf{B}$ and $\mathcal{W}^{(A)}$ of $\mathbf{A}'$ such that*

$$\mathcal{S}\Big(\mathcal{W}^{(B)}\Big) = \mathcal{W}^{(A)},$$

where $\mathcal{S}$ applies the map $x \mapsto x^d$ coordinatewise. (Round constants shift the center of $\mathcal{W}^{(A)}$ and do not affect $\mathcal{S}$-compatibility.)

Definition 7. *A m-round* truncated differential trail *is a sequence of truncated differentials $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^m$ such that for every $i = 1, \ldots, m-1$, $\mathbf{B}_i$ is $\mathcal{S}$-compatible with $\mathbf{A}_{i+1}$.*

Definition 8. *A matrix $\mathbf{B} \in \mathbb{F}^{t\times k}$ is* arity-2 *if*

- *every column of $\mathbf{B}$ has exactly 2 nonzero entries, and*
- *every row of $\mathbf{B}$ has at most 1 nonzero entry.*

The two rows $\ell_j < r_j$ carrying the nonzero entries of column j form a pair*; we write $L = \{\ell_1, \ldots, \ell_k\}$ and $R = \{r_1, \ldots, r_k\}$ for the resulting sets of left and right indices (disjoint by the row condition).*

Definition 9. *An m-round* skipping trail *is an $(m+1)$-round truncated differential trail $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$ such that*

1. *for every i from 1 to m, $\mathbf{B}_i$ is arity-2, and*
2. *there exists a sequence of centers $\{W_i^{(A)}, W_i^{(B)}, W_i^{(C)}\}_{i=1}^{m+1}$ propagating through the trail, i.e. $W_i^{(B)} = M\, W_i^{(A)}$, $W_i^{(C)} = \mathcal{S}(W_i^{(B)})$, and $W_{i+1}^{(A)} = W_i^{(C)} + c_i$ for round constants c_i.*

3.1 A Worked Example: $t = 4$, $M = \mathrm{circ}(2, 1, 1, 1)$

We exhibit a concrete 1-round skipping trail ($m = 1$) for the circulant matrix

$$M = \mathrm{circ}(2, 1, 1, 1) = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}, \quad t = 4,\ k = 1,\ d = 3.$$

Differential matrices. Set $\mathbf{A}_1 = \mathbf{B}_1 = \mathbf{A}_2 = \mathbf{B}_2 = (0, 1, -1, 0)^T \in \mathbb{F}^{4\times 1}$. One checks directly that

$$M\,(0, 1, -1, 0)^T = (0\cdot 2+1\cdot 1+(-1)\cdot 1,\ 1\cdot 2+(-1)\cdot 1,\ 1\cdot 1+(-1)\cdot 2,\ 1\cdot 1+(-1)\cdot 1)^T = (0, 1, -1, 0)^T,$$

so $\mathbf{A}$ is an eigenvector of M with eigenvalue 1. Hence $\mathrm{Im}(M\mathbf{A}_1) = \mathrm{Im}(\mathbf{A}_1) = \mathrm{Im}(\mathbf{B}_1)$, confirming that $(\mathbf{A}_1, \mathbf{B}_1)$ is a valid truncated differential.

Arity-2 check. The single column of $\mathbf{B}_1 = (0, 1, -1, 0)^T$ has exactly two nonzero entries (at positions $\ell = 1$ and $r = 2$ using 0-based indexing), and each row is nonzero in at most one column. Thus $\mathbf{B}_1$ is arity-2 with pair $(\ell, r) = (1, 2)$.

Ratio condition. For $d = 3$ and the pair $(\ell, r) = (1, 2)$:

$$\left(\frac{\mathbf{B}_1[2, 1]}{\mathbf{B}_1[1, 1]}\right)^3 \mathbf{A}_2[1, :] = (-1)^3 \cdot 1 = -1 = \mathbf{A}_2[2, :]. \quad \checkmark$$

Center constraint for $\mathcal{S}$-compatibility. The input coset $\mathcal{W}_1^{(B)} = \{W_0 + X(0, 1, -1, 0)^T : X \in \mathbb{F}\}$ has

$$(W_0[0],\ W_0[1] + X,\ W_0[2] - X,\ W_0[3]).$$

After cubing coordinate-wise (with $u = W_0[1] + X$):

$$\big(W_0[0]^3,\ u^3,\ (W_0[2] - X)^3,\ W_0[3]^3\big).$$

For the image to be a coset of $\mathbf{A}_2 = (0, 1, -1, 0)^T$ we need $(W_0[2] - X)^3 = -u^3$ for all X. Setting $c = W_0[1] + W_0[2]$ gives $W_0[2] - X = c - u$, so $(c - u)^3 = -u^3$ for all u requires $c = 0$, i.e.

$$W_0[1] + W_0[2] = 0.$$

Under this constraint $(W_0[2] - X)^3 = (-u)^3 = -u^3$, and the output coset is

$$\big\{\, (W_0[0]^3,\ W_0[1]^3 + Z,\ -W_0[1]^3 - Z,\ W_0[3]^3) \,:\, Z \in \mathbb{F} \,\big\},$$

which is a coset of $(0, 1, -1, 0)^T = \mathbf{A}_2$. $\mathcal{S}$-compatibility is confirmed.

Summary. The sequence $(\mathbf{A}_1, \mathbf{B}_1) = (\mathbf{A}_2, \mathbf{B}_2) = (0, 1, -1, 0)^T$ forms a valid 1-round skipping trail. Any center W_0 with $W_0[1] + W_0[2] = 0$ satisfies the compatibility condition, leaving the values $W_0[0], W_0[1], W_0[3]$ free. By Theorem 1, the affine line $\{W_0 + X(0, 1, -1, 0)^T\}$ maps through one full round to another affine line, reducing the CICO problem to a univariate system over the remaining rounds (Corollary 1).

With an m-round skipping trail we can algebraically bypass the first m full rounds of the SPN, reducing the effective CICO instance. The following theorem formalizes this and Section 4 derives tight bounds on how large m can be.

4 Theoretical Bounds on Round Skipping

Theorem 1. *Let $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$ be an m-round skipping trail with propagating centers $\{W_i^{(A)}, W_i^{(B)}, W_i^{(C)}\}$. For every $i \in \{1, \ldots, m\}$ and every column j of $\mathbf{B}_i$ with arity-2 pair (ℓ_j, r_j), denote $\rho_{i,j} = \mathbf{B}_i[r_j, j]/\mathbf{B}_i[\ell_j, j]$ and suppose:*

(i) Matrix ratio condition:

$$\rho_{i,j}^d \, \mathbf{A}_{i+1}[\ell_j, :] \;=\; \mathbf{A}_{i+1}[r_j, :]; \tag{1}$$

(ii) State ratio condition:

$$W_i^{(B)}[r_j] \;=\; \rho_{i,j} \, W_i^{(B)}[\ell_j]. \tag{2}$$

Then

$$\big\{ \mathcal{P}^m\big(W_1^{(A)} + \mathbf{A}_1 \cdot \mathbf{X}\big) \big\}_{\mathbf{X} \in \mathbb{F}^k} \;=\; \big\{ W_{m+1}^{(A)} + \mathbf{A}_{m+1} \cdot \mathbf{X}' \big\}_{\mathbf{X}' \in \mathbb{F}^k}. \tag{3}$$

Proof. We prove by induction on $i \in \{1, \ldots, m\}$ that after i full rounds the image of $\{W_1^{(A)} + \mathbf{A}_1\mathbf{X}\}_{\mathbf{X} \in \mathbb{F}^k}$ equals $\{W_{i+1}^{(A)} + \mathbf{A}_{i+1}\mathbf{X}'\}_{\mathbf{X}' \in \mathbb{F}^k}$ for a bijective reparameterization on $\mathbb{F}^k$.

Inductive step. Assume the image after $i-1$ rounds is $\{W_i^{(A)} + \mathbf{A}_i\mathbf{X}\}$ (the base case $i = 1$ is the hypothesis).

Linear layer. Since $(\mathbf{A}_i, \mathbf{B}_i)$ is a truncated differential, $\mathrm{Im}(M\mathbf{A}_i) = \mathrm{Im}(\mathbf{B}_i)$, so $M\mathbf{A}_i = \mathbf{B}_i\mathbf{T}_i$ for an invertible $\mathbf{T}_i \in \mathbb{F}^{k \times k}$. With $\mathbf{Y} = \mathbf{T}_i\mathbf{X}$:

$$\mathcal{L}\big(\{W_i^{(A)} + \mathbf{A}_i\mathbf{X}\}\big) = \{W_i^{(B)} + \mathbf{B}_i\mathbf{Y}\}.$$

S-box layer. Let $v = W_i^{(B)} + \mathbf{B}_i\mathbf{Y}$. Since $\mathbf{B}_i$ is arity-2, for column j with pair (ℓ_j, r_j) and $b_j = \mathbf{B}_i[\ell_j, j]$:

$$v[\ell_j] = W_i^{(B)}[\ell_j] + b_j Y_j, \qquad v[r_j] = W_i^{(B)}[r_j] + \rho_{i,j}\, b_j Y_j.$$

By the state ratio condition (2), $W_i^{(B)}[r_j] = \rho_{i,j} W_i^{(B)}[\ell_j]$, hence $v[r_j] = \rho_{i,j}\, v[\ell_j]$. Setting $Z_j = v[\ell_j]^d - W_i^{(C)}[\ell_j]$ and using $W_i^{(C)}[r_j] = (\rho_{i,j} W_i^{(B)}[\ell_j])^d = \rho_{i,j}^d W_i^{(C)}[\ell_j]$:

$$\mathcal{S}(v)[\ell_j] = W_i^{(C)}[\ell_j] + Z_j, \qquad \mathcal{S}(v)[r_j] = W_i^{(C)}[r_j] + \rho_{i,j}^d Z_j.$$

For any inactive row s (zero row in $\mathbf{B}_i$), $v[s] = W_i^{(B)}[s]$ is constant, so $\mathcal{S}(v)[s] = W_i^{(C)}[s]$. Defining $\mathbf{C}_i \in \mathbb{F}^{t \times k}$ by $\mathbf{C}_i[\ell_j, j] = 1$, $\mathbf{C}_i[r_j, j] = \rho_{i,j}^d$, and zero elsewhere:

$$\mathcal{S}\big(\{W_i^{(B)} + \mathbf{B}_i\mathbf{Y}\}\big) = \{W_i^{(C)} + \mathbf{C}_i\mathbf{Z}\}.$$

Identification with $\mathbf{A}_{i+1}$. By $\mathcal{S}$-compatibility (Definition 3), any inactive row s of $\mathbf{B}_i$ fixes coordinate s under $\mathcal{S}$, so $\mathbf{A}_{i+1}[s, :] = \mathbf{0}$ for all inactive s. Hence $\mathbf{A}_{i+1}$ has support only on $L_i \cup R_i$. The matrix ratio condition (1) then gives $\mathbf{A}_{i+1}[r_j, :] = \rho_{i,j}^d \mathbf{A}_{i+1}[\ell_j, :]$ for each j, which means $\mathrm{Im}(\mathbf{C}_i) = \mathrm{Im}(\mathbf{A}_{i+1})$. Hence $\mathbf{C}_i = \mathbf{A}_{i+1}\mathbf{S}_i$ for an invertible $\mathbf{S}_i \in \mathbb{F}^{k \times k}$; with $\mathbf{X}' = \mathbf{S}_i\mathbf{Z}$:

$$\mathcal{S}\big(\{W_i^{(B)} + \mathbf{B}_i\mathbf{Y}\}\big) = \{W_i^{(C)} + \mathbf{A}_{i+1}\mathbf{X}'\}.$$

Round constant. Since $W_{i+1}^{(A)} = W_i^{(C)} + c_i$, the image after round i is $\{W_{i+1}^{(A)} + \mathbf{A}_{i+1}\mathbf{X}'\}$, completing the induction. Equation (3) follows by taking $i = m$. □

Remark 3 (Cost of finding the center)**.** *Theorem 1 guarantees the output coset* (3) *once a propagating center sequence is known, but does not supply one. Finding valid centers requires solving the system of state ratio conditions* (2)*: for each round* $i \in \{1, \ldots, m\}$ *and each arity-2 pair* $j \in \{1, \ldots, k\}$ *there is one equation* $W_i^{(B)}[r_j] = \rho_{i,j} W_i^{(B)}[\ell_j]$*, giving* mk *equations in total. The round-1 conditions are linear in* $W_1^{(A)}$ *(through* M*); the round-*i *conditions have degree* d^{i-1} *after back-substitution. In the generic case, solving this system consumes* mk *of the* t *input degrees of freedom, leaving* $t - mk$ *free. Of these,* k *coordinates parameterize the differential coset and the remaining* $t - mk - k$ *can be used to satisfy additional input constraints (e.g. CICO fixed inputs).*

Corollary 1 (CICO degree reduction)**.** *Let* $m \leq R_F$ *and suppose there exists an* m*-round skipping trail* $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$ *with valid propagating centers. Let* $I \subseteq \{0, \ldots, t-1\}$ *be the set of rows* s *for which* $\mathbf{A}_1[s, :] = \mathbf{0}$ *(the* inactive positions *of* $\mathbf{A}_1$*). Suppose the* CICO *input constraint fixes exactly the coordinates* $\{W_1^{(A)}[s]\}_{s \in I}$ *(and leaves the remaining* $k = t - |I|$ *coordinates free), so that the input constraint defines a differential coset of* $\mathbf{A}_1$*. Let the remaining* $s = R_F + R_P - m$ *rounds of* $\mathcal{P}$*, viewed as a function of the state at the beginning of round* $m+1$*, be expressed by polynomials of degree at most* D*. Then the* CICO *problem reduces to a system of polynomial equations of degree at most* D *in* k *unknowns.*

Proof. By Theorem 1, the image of the input differential coset under the first m rounds is the differential coset $\{W_{m+1}^{(A)} + \mathbf{A}_{m+1} \cdot \mathbf{X}'\}_{\mathbf{X}' \in \mathbb{F}^k}$. The remaining rounds are a polynomial map of degree D applied to this affine family; substituting the parameterization by $\mathbf{X}'$ turns the CICO output constraints into polynomial equations in $\mathbf{X}'$ of degree at most D. The number of free variables is k, the width of the trail. □

Lemma 1 (Width lower bound from branch number)**.** *Let* $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{m+1}$ *be an* m*-round skipping trail with propagating centers for a linear layer of matrix* M *and branch number* b*. Up to reparameterization we may assume* $M\mathbf{A}_i = \mathbf{B}_i$ *for each* i*. Since every column of* $\mathbf{B}_i$ *is arity-2 (Hamming weight exactly* 2*), the branch number condition applied column-by-column gives* $\mathrm{wt}(\mathbf{A}_i[:, j]) \geq b - 2$ *for every column* j*, so* $\mathbf{A}_i$ *has at most* $t - b + 2$ *zero rows. By* $\mathcal{S}$*-compatibility, the zero rows of* $\mathbf{A}_{i+1}$ *include all* $t - 2k_i$ *zero rows of* $\mathbf{B}_i$*. Therefore*

$$k_i \;\geq\; \frac{b-2}{2} \quad \textit{for all } i \in \{1, \ldots, m\}. \tag{4}$$

Proof. After reparameterizing $\mathbf{A}_i \leftarrow M^{-1}\mathbf{B}_i$, column j of $M\mathbf{A}_i$ equals column j of $\mathbf{B}_i$, which has weight 2. Applying the branch number inequality $\mathrm{wt}(v) + \mathrm{wt}(Mv) \geq b$ with $v = \mathbf{A}_i[:, j]$ gives $\mathrm{wt}(\mathbf{A}_i[:, j]) \geq b - 2$; hence $\mathbf{A}_i$ has at most $t - b + 2$ zero rows. Since a zero row of $\mathbf{B}_i$ at position s means coordinate s is constant throughout the coset $\{W_i^{(B)} + \mathbf{B}_i\mathbf{Y}\}$, $\mathcal{S}$-compatibility forces $\mathbf{A}_{i+1}[s, :] = \mathbf{0}$ as well. Thus $\mathbf{A}_{i+1}$ inherits at least $t - 2k_i$ zero rows from $\mathbf{B}_i$, giving $t - 2k_i \leq t - b + 2$, i.e. $k_i \geq (b-2)/2$. □

Remark 4. *Combining* (4) *with the degrees-of-freedom constraint* $\sum_{i=1}^{m} k_i \leq t$ *(each column of* $\mathbf{B}_i$ *introduces one independent free parameter in the* t*-dimensional state space) yields*

$$\frac{m(b-2)}{2} \;\leq\; \sum_{i=1}^{m} k_i \;\leq\; t,$$

hence $b \leq 2t/m + 2$ *for any skipping trail of length* m*. For an MDS layer (*$b = t + 1$*,* $t \geq 3$*):* $m(t-1) \leq 2t$*, so* $m \leq 2t/(t-1) < 3$*, which rules out skipping* $m \geq 3$ *full rounds.*

Corollary 2 (Per-round active width)**.** *In an m-round skipping trail with linear layer of branch number b, the number of active S-box evaluations in round i is $n_{B_i} = 2k_i \geq b-2$ for every $i \in \{1, \ldots, m\}$. The total number of active S-box evaluations across all m rounds satisfies*

$$\sum_{i=1}^{m} n_{B_i} \;\geq\; m(b-2). \tag{5}$$

Proof. Immediate from Lemma 1: $k_i \geq (b-2)/2$ and $n_{B_i} = 2k_i$. □

Proposition 1. *A skipping trail of length m for a linear layer of branch number b requires*

$$b \;\leq\; \frac{2t}{m} + 2. \tag{6}$$

Proof. By Corollary 2, $\sum_{i=1}^{m} n_{B_i} \geq m(b-2)$. Since each column of $\mathbf{B}_i$ introduces one independent degree of freedom and the state space is t-dimensional, $\sum_{i=1}^{m} k_i \leq t$, i.e. $\sum_{i=1}^{m} n_{B_i} \leq 2t$. Combining:

$$m(b-2) \;\leq\; 2t,$$

which gives (6). □

Proposition 2. *If the linear transformation $\mathcal{L}$ is MDS with $t \geq 3$, then no skipping trail of length $m \geq 3$ exists.*

Proof. For MDS, $b = t+1$. Proposition 1 gives $t+1 \leq 2t/m + 2$, i.e. $m \leq 2t/(t-1)$. For $t \geq 3$ this yields $m \leq 2$. □

5 Concrete Trails

We demonstrate that the bound $m \leq 2$ of Proposition 2 is tight by exhibiting an explicit 2-round skipping trail of width $k = 1$. We work with the smallest nontrivial case $t = 3$, $d = 3$, over the prime field $\mathbb{F}_{11}$; in Section 5.4 we verify the same construction over the KoalaBear field $\mathbb{F}_p$ with $p = 2^{31} - 2^{24} + 1$.

5.1 The MDS Matrix

Let

$$M \;=\; \mathrm{circ}(2,1,1) \;=\; \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

We verify that M is MDS over $\mathbb{F}_{11}$. Its determinant is $\det(M) = 2(4-1) - 1(2-1) + 1(1-2) = 4 \not\equiv 0 \pmod{11}$. All 2×2 submatrix determinants take values in $\{\pm 1, 3\}$, each nonzero modulo 11 (the critical case is $\det\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right) = 3$; all individual entries $1, 2$ are also nonzero). Hence every square submatrix of M is invertible, so M is MDS with branch number $b = t+1 = 4$.

5.2 The Skipping Trail

Set $\mathbf{a} = (0, 1, -1)^T \in \mathbb{F}_{11}^3$. Direct computation gives

$$M\mathbf{a} = \begin{pmatrix} 0+1-1 \\ 0+2-1 \\ 0+1-2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \mathbf{a},$$

so $\mathbf{a}$ is an eigenvector of M with eigenvalue 1. In particular $\mathrm{Im}(M\mathbf{a}) = \mathrm{Im}(\mathbf{a})$, which means $(\mathbf{a}, \mathbf{a})$ is a truncated differential for $\mathcal{L}$.

We set $\mathbf{A}_i = \mathbf{B}_i = \mathbf{a}$ for $i = 1, 2, 3$, giving a candidate 2-round trail $\{(\mathbf{a}, \mathbf{a})\}_{i=1}^3$.

Arity-2 check. The column vector $\mathbf{a}$ has exactly two nonzero entries (rows 1 and 2, 0-indexed), and each row has at most one nonzero. Hence $\mathbf{a}$ is arity-2 (Definition 8) with active pair $(\ell, r) = (1, 2)$.

Matrix ratio condition. The ratio is $\rho = \mathbf{a}[2]/\mathbf{a}[1] = -1$. The matrix ratio condition (1) requires $\rho^d \cdot \mathbf{a}[\ell] = \mathbf{a}[r]$, i.e. $(-1)^3 \cdot 1 = -1 = \mathbf{a}[2]$. This holds for any odd d, in particular $d = 3$. ✓

5.3 Center Constraints and Coset Flow

The trail is determined; it remains to find a center $W_1^{(A)} = (w_0, w_1, w_2)^T$ satisfying Theorem 1. Let $c_1 = (c_{1,0}, c_{1,1}, c_{1,2})^T$ denote the round-1 constant vector.

Deriving the constraints. Set $u_0 = W_1^{(B)}[0] = 2w_0 + w_1 + w_2$ and $v = W_1^{(B)}[1] = w_0 + 2w_1 + w_2$. The state ratio condition (2) for round 1 ($W_1^{(B)}[2] = -W_1^{(B)}[1]$) expands to the linear equation

$$\text{(SR-1)}: \quad 2w_0 + 3(w_1 + w_2) = 0. \tag{7}$$

Since $(-v)^3 = -v^3$ for $d = 3$ odd, the image of $W_1^{(B)}$ under $\mathcal{S}$ is $(u_0^3, v^3, -v^3)^T$. After adding c_1 and applying M, the state ratio condition for round 2 ($W_2^{(B)}[2] = -W_2^{(B)}[1]$) reduces to

$$\text{(SR-2)}: \quad 2u_0^3 + 2c_{1,0} + 3(c_{1,1} + c_{1,2}) = 0. \tag{8}$$

Solving the system. Equation (SR-2) determines u_0 uniquely:

$$u_0 = \left(-\frac{2c_{1,0} + 3(c_{1,1} + c_{1,2})}{2}\right)^{1/3}, \tag{9}$$

where $^{1/3}$ denotes the unique cube root (valid because $\gcd(3, p-1) = 1$; see Section 5.4). From (SR-1) and $u_0 = 2w_0 + w_1 + w_2$ one obtains $w_0 = 3u_0/4$; then $w_1 + w_2 = -u_0/2$ with $w_1 \in \mathbb{F}_p$ free. Thus there exists a *one-parameter family* of valid centers.

Proposition 3. *For any choice of round constants $c_1 \in \mathbb{F}_p^3$, the system* (SR-1)–(SR-2) *has a solution, and the solution set is the affine line $\{(3u_0/4,\ w_1,\ -u_0/2 - w_1)^T : w_1 \in \mathbb{F}_p\}$ with u_0 given by* (9).

Zero round constants. With $c_1 = \mathbf{0}$, equation (9) gives $u_0 = 0$, so $w_0 = 0$ and the center is $(0, c, -c)^T$ for any $c \neq 0$. For $c = 1$ over $\mathbb{F}_{11}$ the coset flow is:

Step	Coset
Input	$\{(0,\ 1+x,\ -1-x)^T : x \in \mathbb{F}_{11}\}$
After $\mathcal{L}_1$	$\{(0,\ 1+x,\ -1-x)^T\}$ (M fixes the eigenspace)
After $\mathcal{S}_1$	$\{(0,\ (1+x)^3,\ -(1+x)^3)^T\}$ (odd d)
After $\mathcal{L}_2$	$\{(0,\ (1+x)^3,\ -(1+x)^3)^T\}$
After $\mathcal{S}_2$	$\{(0,\ (1+x)^9,\ -(1+x)^9)^T\}$

Since $\gcd(9, 10) = 1$ (Fermat: $(1+x)^9 = (1+x)^{-1}$ over $\mathbb{F}_{11}$), the map $x \mapsto (1+x)^9$ is a bijection, confirming that the output is again a coset of $\mathbf{a}$. The 2-round skipping trail is valid.

Proposition 4 (CICO application)**.** *Let $v \in \mathbb{F}_p$ be any prescribed value. The center family of Proposition 3 can satisfy an additional input constraint $W_1^{(A)}[1] = v$ by choosing $w_1 = v$ with coset shift $x = 0$, yielding the unique trail-compatible input*

$$W^* \;=\; \left(\tfrac{3u_0}{4},\ v,\ -\tfrac{u_0}{2} - v\right)^T, \tag{10}$$

where u_0 is determined by (9) from the round constants. For $v = 0$ in particular, $W^* = (3u_0/4,\ 0,\ -u_0/2)^T$ is fully fixed by the round constants alone. By Corollary 1, the 2-round image $\Pi_2(W^*)$ lies in the output coset of $\mathbf{a}$ and can be evaluated directly, reducing the CICO problem on the full permutation to a system on the remaining $R_F - 2$ full rounds and R_P partial rounds with no free parameters contributed by the first two rounds.

Proof. Set $w_1 = v$ and $x = 0$ in the center family; equation (10) follows immediately. The state ratio conditions (SR-1)–(SR-2) are satisfied by construction (Proposition 3), so Theorem 1 applies and Corollary 1 gives the degree reduction. □

5.4 Extension to KoalaBear

The field $\mathbb{F}_p$ with $p = 2^{31} - 2^{24} + 1 = 2130706433$ (KoalaBear) satisfies:

1. $\det(M) = 4 \neq 0 \pmod p$ (since $p > 4$) and $3 \neq 0 \pmod p$, so all 2×2 minor values $\{1, 2, 3, 4\}$ are nonzero. Hence $M = \mathrm{circ}(2, 1, 1)$ is MDS over $\mathbb{F}_p$. ✓
2. $p - 1 = 2^{24} \cdot 127$. Since $127 = 42 \cdot 3 + 1$, we have $3 \nmid (p-1)$, so $\gcd(3, p-1) = 1$ and $x \mapsto x^3$ is a bijection on $\mathbb{F}_p$. ✓

Both conditions being satisfied, the trail $\mathbf{A}_i = \mathbf{B}_i = \mathbf{a}$ and Proposition 3 carry over verbatim to KoalaBear. For concrete Poseidon1-KoalaBear round constants, u_0 is computed from (9) as the unique cube root of the right-hand side.

Remark 5. *The circulant MDS matrix used in the actual Poseidon1-KoalaBear bounty has dimension $t = 16$, not $t = 3$. The present section demonstrates tightness of Proposition 2 using the simplest possible MDS instance; explicit $m = 2$ trails for $t = 16$ are left for future work.*

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